

Network Reduction Technique: Source Transformation

High-Level Overview

Source Transformation is a fundamental network reduction technique used in circuit analysis to simplify complex electrical circuits. It allows us to substitute a practical voltage source with an equivalent practical current source, or vice versa, without altering the behavior of the rest of the circuit external to the transformation terminals.

Core Concepts & Fundamentals

- Electric Circuit:** An interconnection of electrical components containing at least one closed loop through which electric current can flow continuously.
- Circuit Elements:**
 - Active Elements:** Components capable of generating electrical energy (e.g., independent voltage sources, independent current sources).
 - Passive Elements (Sinks):** Components that absorb or dissipate energy (e.g., Resistors R , Inductors L , Capacitors C).
- Practical vs. Ideal Sources:**
 - Ideal Voltage Source:** Maintains a constant voltage across its terminals regardless of the current drawn.
 - Practical Voltage Source:** Modeled as an ideal voltage source V_S connected in **series** with an internal resistance R_S .
 - Ideal Current Source:** Delivers a constant current regardless of the terminal voltage.
 - Practical Current Source:** Modeled as an ideal current source I_S connected in **parallel** with an internal resistance R_S .

Intuition Behind Equivalence

Two circuit models are **equivalent** if they produce the exact same voltage across and current through any load connected between terminals A and B .

Diagram: Source Transformation Equivalence

Description: The left portion shows a practical voltage source consisting of an ideal voltage source V_S in series with a resistor R_S , connected to terminals A and B . An arrow points to the right portion, which shows an equivalent practical current source consisting of an ideal current source I_S directed upward, connected in parallel with the same resistor R_S between terminals A and B .

Conversion Rules & Mathematical Derivation

To transform between a practical voltage source and a practical current source across terminals A and B :

Transformation Formulas

- Voltage Source to Current Source Transformation:**
 - Keep the internal resistance value the same: R_S .
 - Place R_S in **parallel** with the new current source.
 - Calculate the current source value using Ohm's Law:

$$I_S = \frac{V_S}{R_S}$$

- Current Source to Voltage Source Transformation:**
 - Keep the internal resistance value the same: R_S .
 - Place R_S in **series** with the new voltage source.
 - Calculate the voltage source value using Ohm's Law:

$$V_S = I_S \cdot R_S$$

Rule for Source Polarity / Current Direction

- The arrow of the current source I_S must point toward the terminal corresponding to the positive (+) terminal of the voltage source V_S .

Step-by-Step Worked Examples

Illustration 1: Circuit Reduction using Source Transformation

Problem Statement:

Determine the current through and the power consumed by the $15\ \Omega$ resistor in the circuit below.

Diagram: Illustration 1 Original Circuit

Description: Original circuit diagram. Leftmost branch: a $100\ \Omega$ resistor in parallel with a $20\ \text{A}$ current source (pointing upward). Middle top: a $6\ \Omega$ resistor in series. Middle branch: a $6\ \Omega$ resistor in parallel. Right top: a $6\ \Omega$ resistor in series. Right middle branch: a branch containing a $50\ \text{V}$ voltage source (positive terminal on top) in series with a $0.5\ \Omega$ resistor. Rightmost branch: the $15\ \Omega$ target load resistor in parallel.

Step 1: Simplify the Leftmost Parallel Branch

Look at the leftmost part of the circuit: a $20\ \text{A}$ current source is in parallel with a $100\ \Omega$ resistor.

Convert this practical current source to an equivalent voltage source:

- **Equivalent Voltage Source Value (V_{S1}):**

$$V_{S1} = I \cdot R = 20\ \text{A} \times 100\ \Omega = 2000\ \text{V}$$

- **Series Resistance:** $100\ \Omega$

The $100\ \Omega$ resistor is now in series with the adjacent $6\ \Omega$ resistor.

- **Combined Series Resistance (R_{s1}):**

$$R_{s1} = 100\ \Omega + 6\ \Omega = 106\ \Omega$$

Now, transform this $2000\ \text{V}$ source in series with $106\ \Omega$ back to a current source:

- **Equivalent Current Source Value (I_{S1}):**

$$I_{S1} = \frac{2000\ \text{V}}{106\ \Omega} = \frac{1000}{53}\ \text{A} \approx 18.868\ \text{A}$$

- **Parallel Resistance:** $106\ \Omega$
-

Step 2: Combine Parallel Resistances in the Middle Stage

The parallel $106\ \Omega$ resistor is now in parallel with the middle vertical $6\ \Omega$ resistor:

- **Equivalent Parallel Resistance (R_{p1}):**

$$R_{p1} = 106\ \Omega \parallel 6\ \Omega = \frac{106 \times 6}{106 + 6} = \frac{636}{112} = \frac{159}{28}\ \Omega \approx 5.6786\ \Omega$$

Now, transform this parallel combination (current source $I_{S1} = \frac{1000}{53}\ \text{A}$ in parallel with $R_{p1} = \frac{159}{28}\ \Omega$) back into a voltage source:

- **Equivalent Voltage Source Value (V_{S2}):**

$$V_{S2} = I_{S1} \cdot R_{p1} = \left(\frac{1000}{53} \right) \times \left(\frac{159}{28} \right) = \frac{1000 \times 3}{28} = \frac{3000}{28} = \frac{750}{7}\ \text{V} \approx 107.143\ \text{V}$$

- **Series Resistance:** $R_{p1} = \frac{159}{28}\ \Omega$
-

Step 3: Combine Series Resistances Leading to the Output Stage

This new voltage source ($V_{S2} = \frac{750}{7}$ V) is in series with its internal resistance $R_{p1} = \frac{159}{28} \Omega$ and the top middle 6Ω resistor.

- **Total Series Resistance (R_{s2}):**

$$R_{s2} = \frac{159}{28} + 6 = \frac{159 + 168}{28} = \frac{327}{28} \Omega \approx 11.6786 \Omega$$

Now, transform this branch ($V_{S2} = \frac{750}{7}$ V in series with $R_{s2} = \frac{327}{28} \Omega$) back into a current source:

- **Equivalent Current Source Value (I_{S2}):**

$$I_{S2} = \frac{V_{S2}}{R_{s2}} = \frac{\frac{750}{7}}{\frac{327}{28}} = \frac{750}{7} \times \frac{28}{327} = \frac{750 \times 4}{327} = \frac{3000}{327} = \frac{1000}{109} \text{ A} \approx 9.1743 \text{ A}$$

- **Parallel Resistance:** $R_{p2} = \frac{327}{28} \Omega$
-

Step 4: Handle the Right-Hand Branch with 50 V Source

The right-hand branch has a 50 V voltage source in series with a 0.5Ω resistor.

Convert this into an equivalent current source:

- **Equivalent Current Source Value (I_{S3}):**

$$I_{S3} = \frac{50 \text{ V}}{0.5 \Omega} = 100 \text{ A}$$

- **Parallel Resistance:** 0.5Ω
-

Step 5: Combine All Parallel Sources and Resistances Connected to Load

Now, connected in parallel across the 15Ω load resistor, we have:

1. Current source $I_{S2} = \frac{1000}{109} \text{ A}$ (pointing upward)
2. Resistor $R_{p2} = \frac{327}{28} \Omega$
3. Current source $I_{S3} = 100 \text{ A}$ (pointing upward)
4. Resistor $R_{p3} = 0.5 \Omega$
5. Load resistor $R_L = 15 \Omega$

Combine the two parallel current sources:

$$I_{total} = I_{S2} + I_{S3} = \frac{1000}{109} + 100 = \frac{11900}{109} \text{ A} \approx 109.1743 \text{ A}$$

Combine the internal parallel resistances (R_{p2} and R_{p3}):

$$\frac{1}{R_{eq_int}} = \frac{1}{R_{p2}} + \frac{1}{R_{p3}} = \frac{28}{327} + \frac{1}{0.5} = \frac{28}{327} + 2 = \frac{28 + 654}{327} = \frac{682}{327} \Omega^{-1}$$

$$R_{eq_int} = \frac{327}{682} \Omega \approx 0.4795 \Omega$$

Step 6: Final Current and Power Calculation

Using the **Current Divider Rule** for the total parallel current I_{total} split between R_{eq_int} and $R_L = 15 \Omega$:

$$I_{15\Omega} = I_{total} \times \frac{R_{eq_int}}{R_{eq_int} + 15}$$

Substitute values:

$$I_{15\Omega} = \left(\frac{11900}{109} \right) \times \frac{\frac{327}{682}}{\frac{327}{682} + 15} = \left(\frac{11900}{109} \right) \times \frac{327}{327 + 10230} = \left(\frac{11900}{109} \right) \times \frac{327}{10557}$$

Simplifying:

$$I_{15\Omega} = \frac{11900 \times 3}{10557} = \frac{35700}{10557} \approx 3.3816 \text{ A}$$

Power Consumed by the 15 Ω Resistor ($P_{15\Omega}$):

$$P_{15\Omega} = (I_{15\Omega})^2 \cdot R_L = (3.3816)^2 \times 15 \approx 11.435 \times 15 \approx 171.53 \text{ W}$$

Illustration 2: Reduction to a Practical Voltage Source Across a Load

Problem Statement:

1. Simplify the given circuit into a practical voltage source connected across the load resistor R_L .
2. Calculate the voltage drop across the load resistance when $R_L = 10 \Omega$.

 Diagram: Illustration 2 Original Circuit

Description: Circuit diagram containing two parallel practical voltage sources on the left: 24 V in series with 0.1 Ω , and 24 V in series with 0.15 Ω . These connect to a node leading to a parallel combination of 40 Ω and 60 Ω in series with a 6 Ω resistor, a diagonal 100 Ω resistor to ground, and an output load R_L .

Step 1: Convert Both Parallel Voltage Sources to Current Sources

- **Source 1** (24 V, 0.1 Ω):

$$I_{S1} = \frac{24 \text{ V}}{0.1 \Omega} = 240 \text{ A}, \quad R_{p1} = 0.1 \Omega$$

- **Source 2** (24 V, 0.15 Ω):

$$I_{S2} = \frac{24 \text{ V}}{0.15 \Omega} = 160 \text{ A}, \quad R_{p2} = 0.15 \Omega$$

Step 2: Combine the Parallel Current Sources and Resistances

- **Total Parallel Current Source (I_{S_comb}):**

$$I_{S_comb} = 240 \text{ A} + 160 \text{ A} = 400 \text{ A}$$

- **Equivalent Parallel Resistance (R_{p_comb}):**

$$R_{p_comb} = 0.1 \Omega \parallel 0.15 \Omega = \frac{0.1 \times 0.15}{0.1 + 0.15} = \frac{0.015}{0.25} = 0.06 \Omega$$

Step 3: Transform Back to a Single Equivalent Voltage Source

- **Combined Voltage Source (V_{S_comb}):**

$$V_{S_comb} = I_{S_comb} \cdot R_{p_comb} = 400 \text{ A} \times 0.06 \Omega = 24 \text{ V}$$

- **Series Resistance:** 0.06 Ω

Step 4: Simplify the Resistor Network Towards Load R_L

1. **Parallel Resistors** (40 $\Omega \parallel 60 \Omega$):

$$R_{40\parallel 60} = \frac{40 \times 60}{40 + 60} = \frac{2400}{100} = 24 \Omega$$

2. Diagonal 100 Ω Resistor:

The 24 V source with series resistance 0.06 Ω is connected across the 100 Ω resistor.

Convert the 24 V source with 0.06 Ω to a current source:

$$I = \frac{24 \text{ V}}{0.06 \Omega} = 400 \text{ A}, \quad R = 0.06 \Omega$$

Combine parallel resistors 0.06 $\Omega \parallel 100 \Omega$:

$$R' = \frac{0.06 \times 100}{0.06 + 100} = \frac{6}{100.06} \approx 0.059964 \Omega$$

Convert back to a voltage source:

$$V' = 400 \text{ A} \times 0.059964 \Omega \approx 23.9856 \text{ V} \quad (\approx 24 \text{ V})$$

Internal series resistance: $R_{eq_src} \approx 0.06 \Omega$.

3. Total Series Resistance leading to R_L :

$$R_{th} = R_{eq_src} + 6 \Omega + R_{40\parallel 60} = 0.06 \Omega + 6 \Omega + 24 \Omega = 30.06 \Omega$$

Step 5: Final Answers for Part (a) and Part (b)

(a) Practical Voltage Source Equivalent:

- **Equivalent Voltage Source (V_{eq}):** 24 V (or 23.986 V)
- **Equivalent Series Resistance (R_{eq}):** 30.06 Ω

(b) Voltage Drop Across Load $R_L = 10 \Omega$:

Using the Voltage Divider Rule:

$$V_{R_L} = V_{eq} \times \left(\frac{R_L}{R_{eq} + R_L} \right) = 24 \text{ V} \times \left(\frac{10 \Omega}{30.06 \Omega + 10 \Omega} \right) = 24 \times \frac{10}{40.06} \approx 5.991 \text{ V}$$

Illustration 3: Power Dissipation in a Specific Resistor

Problem Statement:

In the circuit shown, find the amount of power dissipated in the 3 Ω resistor.

 Diagram: Illustration 3 Original Circuit

Description: Circuit diagram with multiple branches. Left side has a top 24 V source, middle loop with a 6 V voltage source in series with a 6 Ω resistor, bottom 12 V source. Middle branch contains a 3 Ω resistor. Top branch has a 6 Ω resistor, rightmost branch has a 6 Ω resistor.

Step 1: Identify Terminal Connections and Loop Voltages

Let us define Node Voltages to easily solve for the voltage across the target 3 Ω resistor:

1. Define the reference node (Ground, 0 V) at the node between the 24 V source and the 12 V source (middle-left node).
2. **Node 1 (Top-Left, above 24 V source):**

$$V_{top_left} = +24 \text{ V}$$

3. **Node 2 (Bottom-Left, below 12 V source):**

$$V_{bottom_left} = -12 \text{ V}$$

4. Middle-Left Node (between sources):

$$V_{mid_left} = 0 \text{ V}$$

Notice that the bottom terminal of the 3Ω resistor is directly connected to $V_{mid_left} = 0 \text{ V}$.

Let V_A be the voltage at the top terminal of the 3Ω resistor (top-middle node).

Step 2: Apply Nodal Analysis at Node A

Node A connects three branches:

1. Branch to the left (through 6Ω resistor to $+24 \text{ V}$):

$$I_1 = \frac{V_A - 24}{6}$$

2. Branch downwards through the 3Ω resistor to 0 V :

$$I_2 = \frac{V_A - 0}{3}$$

3. Branch downwards through the bottom loop (contains 6 V source and 6Ω resistor) and rightmost 6Ω resistor connected to bottom node (-12 V):

The path from $V_{mid_left} = 0 \text{ V}$ goes through 6 V source (positive facing right), so node before 6Ω is $+6 \text{ V}$.

The rightmost branch connects to -12 V through a 6Ω resistor.

Applying Kirchhoff's Current Law (KCL) at Node A ($\sum I_{out} = 0$):

$$\frac{V_A - 24}{6} + \frac{V_A - 0}{3} + \frac{V_A - 6}{6} + \frac{V_A - (-12)}{6} = 0$$

Multiply the entire equation by 6 to clear fractions:

$$(V_A - 24) + 2(V_A) + (V_A - 6) + (V_A + 12) = 0$$

Combine like terms:

$$(1 + 2 + 1 + 1)V_A - 24 - 6 + 12 = 0$$

$$5V_A - 18 = 0$$

$$5V_A = 18 \implies V_A = \frac{18}{5} = 3.6 \text{ V}$$

Step 3: Calculate Power Dissipating in the 3Ω Resistor

The voltage across the 3Ω resistor is:

$$V_{3\Omega} = V_A - 0 \text{ V} = 3.6 \text{ V}$$

Using the power formula $P = \frac{V^2}{R}$:

$$P_{3\Omega} = \frac{(V_{3\Omega})^2}{R} = \frac{(3.6)^2}{3} = \frac{12.96}{3} = 4.32 \text{ W}$$

Summary Reference Table

Feature	Practical Voltage Source	Practical Current Source
Ideal Component	Ideal Voltage Source (V_S)	Ideal Current Source (I_S)
Internal Resistance Connection	Series (R_S)	Parallel (R_S)
Conversion Equation	$V_S = I_S \cdot R_S$	$I_S = \frac{V_S}{R_S}$
Resistance Value R_S	Unchanged (R_S)	Unchanged (R_S)
Polarity / Direction Rule	Positive terminal (+)	Current arrow ($\uparrow$) points to (+)